

MRITUNJOY HALDER

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 Mritunjoy Halder

EXPERIENCE

Researcher

TCS Research Labs

 July 2023 – Present

 Kolkata

- Bridging generative modeling and 3D vision to create simulators for AR/VR-driven robot training.
- Developed an algorithm to address the lack of 3D datasets in current SOTA methods.
- Enabled businesses to generate 3D environments and objects from text/audio prompts.

Research Intern

TCS Research Labs

 March 2022 – May 2022

 Remote

- Proposed a spatio-temporal CNN, inspired by brain cells, for anomaly detection in surveillance robots.
- Achieved 91% accuracy on a benchmark dataset and validated on a Double Robot in real-world scenarios.
- Granted patent; accepted for oral presentation at IEEE IJCNN.

Research Intern

NIT Agartala

 June 2021 – May 2022

 Remote

- Designed a hybrid classical and learning-based model for real-time image defogging.
- Developed a fog dilution and removal model, achieving 92% SSIM with 0.1s processing time.

Research Intern

NIT Hamirpur

 June 2021 – May 2022

 Remote

- Developed a transformer based system for sex, accent, and emotion recognition from speech.
- Achieved 94.62% (sex), 97.37% (accent), and 99.84% (emotion) recognition accuracy.

PUBLICATIONS

Published Journal **A Transmission Model based Deep Neural Network for Image Dehazing** at *Multimedia Tools and Application, Springer*

Published Conference Paper **Anomalous Activity Detection from Ego View Camera of Surveillance Robots** at *IEEE International Joint Conference on Neural Networks (IJCNN)*

Published Conference Paper **A Framework for Sex Identification, Accent and Emotion Recognition from Speech Samples** at *International Conference on Computing Communication and Networking Technologies (ICCCNT)*

Published Journal **Multi-feature based hazy image classification for vision enhancement** at *Procedia Computer Science, Elsevier*

Published Journal **A deep learning model to detect foggy images for vision enhancement** at *The Imaging Science Journal, Taylor and Francis*

Published Book Chapter **Dehazing and vision enhancement: challenges and future scope** at *IET Intelligent Multimedia Processing and Computer Vision*

PATENTS

Part of granted US Patent **Anomalous activity detection for mobile surveillance robots**

Part of published Indian Patent **System and Method For Dehaizng Real Time Dynamic Degraded Scenes Using Deep Residual Neural Networks**

WORK UNDER REVIEW

Paper titled **FAV3R: Fast and Accurate 3D VR Sketch to 3D Shape Retrieval** is under peer review at *IEEE ISMAR (IEEE Transactions on Visualization and Computer Graphics)*

PROJECTS

Improved Diagnosis on Low Resolution Medical Images

B.Tech Thesis

 Dr. Santi Prasad Maity

- Proposed a dual GAN framework for enhancing low-resolution medical images, ensuring better diagnosis.
- Introduced a novel loss function to improve the region of interest (ROI) reconstruction.
- Achieved 91% accuracy, outperforming state-of-the-art methods.

Cartoon Emotion Recognition

PAN India Hackathon Revelation'23

 IIEST Shibpur

- Developed a transformer based model for sentiment classification using a custom cartoon dataset.
- Leveraged synthetic images to train and fine-tune the model for emotion recognition.
- Achieved 95% accuracy, surpassing existing sentiment classification methods.

EDUCATION

B.Tech in Information Technology

Indian Institute of Engineering Science and Technology, Shibpur

 2019 – 2023

 CGPA: 9.1

Higher Secondary in Science

Jadavpur Vidyapith

 2017 – 2019

 90.2 %

Secondary from West Bengal Board

Jadavpur Vidyapith

 2011 – 2017

 88.28 %

QUANTITATIVE RESEARCH

GenAI (VLM/LLM/3D)	
Computer Vision	
Computer Graphics	
Image Processing	
Deep Learning	
Machine Learning	

SKILLS

Python	
C/C++	
Blender(BPY)	
MATLAB	
LaTeX	

LANGUAGE

English	
Bengali	
Hindi	